CipherScope: LLM-Assisted Hybrid Machine Learning Classification of Symmetric Encryption Algorithms from Ciphertext Statistical Features

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***Abstract*—Identifying the encryption algorithm applied to a given ciphertext—without access to the key or original plaintext—is a fundamental challenge in cryptanalysis, digital forensics, and network security auditing. This paper presents CipherScope, an LLM-assisted hybrid machine learning system that classifies ciphertext produced by four algorithm families: AES, DES/3DES (merged as DES_FAMILY), RC4, and Affine. A synthetic dataset of 25,000 balanced, labeled samples is generated from random fixed-length plaintexts encrypted across five ciphers. Each sample is encoded as a 532-dimensional statistical feature vector comprising Shannon entropy, byte-frequency distribution, chi-square uniformity statistic, byte-level bigram transition probabilities, and block-level entropy moments. An XGBoost classifier trained on this representation achieves 100% accuracy on a held-out test set of 5,000 samples with a perfect confusion matrix. We critically examine this result, attributing it to dataset regularities—fixed plaintext length, ECB mode throughout, and stark distributional differences between modern and classical ciphers—rather than general cryptanalytic capability. The system is further augmented by a Large Language Model (LLM) layer that provides natural-language explanations of predictions. Key limitations including overfitting risk, inability to distinguish encryption modes, statistical collapse of DES and 3DES, and expected performance degradation on real-world ciphertext are analyzed in depth. A publicly accessible web demonstration is available at https://cipherscope.onrender.com/.**

***Index Terms*—ciphertext classification, symmetric encryption, XGBoost, statistical cryptanalysis, machine learning, Shannon entropy, byte frequency analysis, AES, DES, RC4, Affine cipher, LLM, hybrid AI**

I. Introduction

Symmetric encryption algorithms underpin the security of modern communication systems, from TLS handshakes to filesystem encryption. When encrypted data is intercepted during forensic investigations or network monitoring, a critical question arises: *which algorithm was used?* Classical approaches rely on metadata—file headers, protocol identifiers, or filesystem annotations—which are frequently unavailable in adversarial scenarios. In such cases, the *ciphertext bytes themselves* are the only available evidence.

The foundational question of this work is: *can statistical properties of a raw byte sequence serve as reliable fingerprints for the algorithm that produced it?* By design, modern symmetric ciphers produce output computationally indistinguishable from random noise. Yet implementation artifacts persist: block sizes, padding schemes, and cipher-specific biases may, in combination, be learnable by a statistical model. Gradient-boosted ensembles [1] excel on structured tabular data, while LLMs [2] enable hybrid systems combining classification precision with natural-language interpretability.

CipherScope trains an XGBoost classifier on statistical features from hex-encoded ciphertext, augments predictions with LLM-generated explanations via the Anthropic Claude API, and exposes the pipeline through an interactive web application. On a controlled synthetic benchmark it achieves perfect accuracy—a result we treat as a proof-of-concept whose precisely characterized limitations define the path toward practical deployment.

Contributions:

1) A reproducible synthetic dataset pipeline spanning five cipher families.
2) A compact 532-dimensional feature representation capturing entropy, byte-frequency, bigram, and block-structural properties.
3) An XGBoost classifier achieving perfect benchmark accuracy with rigorous analysis of *why* this holds under controlled conditions.

4) A critical discussion of generalization limitations and dataset biases.
5) A deployed hybrid LLM+ML web application with real-time natural-language explanation.

II. Related Work

Statistical cryptanalysis. Fluhrer, Mantin, and Shamir [3] showed that RC4's key-scheduling algorithm introduces keystream biases enabling partial key recovery, establishing that even "secure" stream ciphers can leak statistical information and formalizing the concept of a *distinguisher*.

ML for cipher identification. Dileep and Sekhar [4] applied SVMs with frequency-based features to distinguish AES, DES, and Blowfish on synthetic data. Leierzopf et al. [5] evaluated multiple models on engineered statistical features, highlighting the importance of feature representation. Chou et al. [6] demonstrated that ciphertext statistical properties can differentiate encryption schemes. These works validate statistical ML approaches under controlled conditions while acknowledging real-world generalization as an open problem.

Deep learning on byte sequences. Anderson and McGrew [7] classified encrypted network traffic using RNNs; Wang et al. [8] applied CNNs to encrypted file classification—reducing reliance on hand-engineered features at the cost of interpretability.

LLM-augmented security tools. GPT-4 and Claude have been applied to vulnerability explanation [9] and CTF problem solving. CipherScope extends this trend by using the LLM solely as an explanation engine over a purpose-built statistical classifier, avoiding hallucination risks in security-sensitive predictions.

III. Methodology

Algorithm 1 summarizes the data generation process. All encryption is performed using the `pycryptodome` library [10]. For each cipher, 5,000 independent samples are generated using fresh random keys and random plaintexts drawn uniformly from the 62-character alphanumeric alphabet.

Algorithm 1 Synthetic Dataset Generation

```
for each algorithm A ∈ {AES, DES, 3DES, RC4, Affine} do
    for i = 1 to 5000 do
        Sample P ← Uniform([A-Za-z0-9]^64)
        Sample fresh random key K for algorithm A
        Compute C ← Encrypt_A(P, K)
        Append (hex(C), A) to dataset D
    end for
end for
Shuffle D; replace DES, 3DES labels → DES_FAMILY
Split D: 80% train, 20% test (stratified)
```

Table I summarizes the encryption parameters. The use of ECB mode throughout is a deliberate simplification for this initial study; its implications for learnability are analyzed in Section V.

TABLE I
Encryption Parameters by Algorithm

Algorithm	Key Size	Mode / Type	CT Bytes
AES	128-bit	ECB + PKCS#7	80
DES	64-bit	ECB + PKCS#7	72
3DES	192-bit	ECB + PKCS#7	72
RC4	128-bit	Stream cipher	64
Affine	a=5, b=8	Classical subst.	64*

*Affine preserves plaintext length; non-alpha bytes pass through unchanged.

A. Feature Engineering

Each hex ciphertext is decoded to a byte sequence $\mathbf{b} = (b_1, \ldots, b_n) \in \{0, \ldots, 255\}^n$. The feature vector $\mathbf{x} \in \mathbb{R}^{532}$ is formed by concatenating four groups described below.

1) Scalar Global Statistics (4 dimensions): **Shannon Entropy** quantifies information content per byte:

$$H(\mathbf{b}) = -\sum_{v=0}^{255} p_v \log_2 p_v, \quad p_v = \frac{\text{count}(v)}{n} \tag{1}$$

Modern ciphers target $H \approx 8$ bits/byte. The Affine cipher, limited to the 26-letter lowercase alphabet, achieves $H \leq \log_2 26 \approx 4.70$ bits/byte.

Byte Variance measures spread around the uniform distribution:

$$\sigma^2 = \frac{1}{256} \sum_{v=0}^{255} \left(p_v - \tfrac{1}{256}\right)^2 \tag{2}$$

Chi-Square Statistic formally tests departure from byte-level uniformity:

$$\chi^2 = \sum_{v=0}^{255} \frac{(f_v - n/256)^2}{n/256} \tag{3}$$

where f_v is the raw count of byte v. Known RC4 keystream biases [3] produce systematic deviations from χ^2_{255}.

Ciphertext Length (bytes) is the dominant discriminator under fixed 64-character plaintexts with deterministic PKCS#7 padding: AES $\rightarrow$ 80 B, DES/3DES $\rightarrow$ 72 B, RC4/Affine $\rightarrow$ 64 B.

2) Byte Frequency Histogram (256 dimensions): The normalized 256-bin histogram $\mathbf{f} = (p_0, \ldots, p_{255})$ encodes the full marginal distribution. This group alone separates Affine ciphertext (values concentrated in 97–122, ASCII `a–z`) from modern cipher output (approximately uniform over 0–255).

3) Bigram Transition Matrix (256 dimensions): A 16×16 matrix $\mathbf{T}$ captures sequential byte dependencies via lower-nibble bigrams:

$$T_{ij} = \sum_{k=1}^{n-1} \mathbf{1}[b_k \bmod 16 = i \ \wedge \ b_{k+1} \bmod 16 = j] \quad (4)$$

After row-normalization and flattening to 256 dimensions, this captures short-range sequential structure. ECB-mode block ciphers exhibit near-uniform inter-block transitions; stream ciphers and classical ciphers may exhibit different patterns under structured plaintexts.

4) Block-Level Entropy Features (3 dimensions): The ciphertext is partitioned into non-overlapping 8-byte blocks $\{\beta_1, \ldots, \beta_m\}$. Three statistics are extracted:

- $\mu_H = \frac{1}{m}\sum_j H(\beta_j)$: mean block entropy
- $\sigma_H^2 = \frac{1}{m}\sum_j (H(\beta_j) - \mu_H)^2$: variance of block entropy
- $\rho = |\{\beta_j\}|/m$: unique-block ratio (detects ECB repetition)

Table II summarizes the complete feature vector.

TABLE II
FEATURE VECTOR COMPOSITION ($|\mathbf{x}| = 532$)

Feature Group	Dim.	Description
Entropy, Variance, χ^2, Length	4	Global scalar statistics
Byte frequency histogram	256	Normalized 256-bin distribution
Transition matrix	256	Lower-nibble bigram probabilities
Block entropy statistics	3	Mean, variance, unique-block ratio
Total	**532**	

B. XGBoost Classifier

XGBoost [1] is selected for its strong empirical performance on structured tabular data, built-in regularization via shrinkage and column subsampling, and native multi-class support via the softmax objective. Hyperparameters: 600 estimators, maximum depth 10, learning rate η=0.03, row subsample 0.9. Labels are integer-encoded: AFFINE→0, AES→1, DES_FAMILY→2, RC4→3. The dataset is split 80/20 with stratified sampling and `random_state=42`.

C. LLM Integration Architecture

After the XGBoost model returns a predicted class and four-class probability vector, the extracted scalar features (entropy, χ^2, byte variance, length, block uniqueness ratio) are formatted into a structured prompt submitted to the Anthropic Claude API (`claude-sonnet`). The LLM generates a 2–4 sentence natural-language explanation of the prediction, citing specific feature values. Critically, the LLM is never asked to perform the classification—it serves solely as an explanation engine, separating classification responsibility (XGBoost) from interpretability (LLM) and avoiding hallucination risks in the security-sensitive prediction path.

IV. EXPERIMENTAL SETUP

A. Implementation Environment

All experiments run in Python 3.10. Key library versions: `xgboost` 2.0.3, `scikit-learn` 1.4.0, `pycryptodome` 3.20.0, `numpy` 1.26.4, `pandas` 2.1.4. The web application backend uses Flask 3.0, deployed on Render.com (512 MB RAM). Dataset generation completes in ≈45 seconds; model training completes in under 5 minutes on a single CPU core. The serialized XGBoost model is loaded at application startup via `joblib`.

B. Dataset Statistics

Table III shows the post-merge dataset composition with train/test breakdown. The DES_FAMILY class is twice the size of others (10,000 samples, reflecting 5,000 DES + 5,000 3DES), introducing mild class imbalance. Given the perfect separability demonstrated in results, no resampling is required.

TABLE III
DATASET COMPOSITION AFTER DES/3DES MERGING

Label	Train	Test	Total	CT Bytes
AES	4,000	1,000	5,000	80
DES_FAMILY	8,000	2,000	10,000	72
RC4	4,000	1,000	5,000	64
AFFINE	4,000	1,000	5,000	64
Total	**20,000**	**5,000**	**25,000**	—

C. Evaluation Protocol

Performance is reported using per-class precision, recall, F1-score, and the full 4×4 confusion matrix,

computed exclusively on the held-out test split. Macro-averaged and weighted-averaged metrics are also reported. Training-set metrics are deliberately omitted to prevent conflation of train and test performance.

V. Results and Discussion

A. Classification Performance

Table IV presents the full classification report. The XGBoost model achieves perfect precision, recall, and F1-score (1.00) across all four classes. The confusion matrix is a 4×4 diagonal matrix with zero off-diagonal entries, confirming zero misclassifications on 5,000 test samples.

TABLE IV
Classification Report on Test Set ($N = 5{,}000$)

Class	Precision	Recall	F1-Score	Support
AES	1.00	1.00	1.00	1,024
AFFINE	1.00	1.00	1.00	950
DES_FAMILY	1.00	1.00	1.00	2,014
RC4	1.00	1.00	1.00	1,012
Accuracy			1.00	5,000
Macro avg	1.00	1.00	1.00	5,000
Weighted avg	1.00	1.00	1.00	5,000

Confusion Matrix.

$$\begin{pmatrix} 1024 & 0 & 0 & 0 \\ 0 & 950 & 0 & 0 \\ 0 & 0 & 2014 & 0 \\ 0 & 0 & 0 & 1012 \end{pmatrix}$$

Rows/columns ordered: AES, AFFINE, DES_FAMILY, RC4.

B. Analysis of Perfect Accuracy: Four Separating Factors

The perfect result warrants careful scrutiny. We identify four orthogonal factors, each individually sufficient to separate at least two classes, explaining why a combined 532-feature model achieves zero errors.

Factor 1: Ciphertext Length as a Hard Separator. For fixed 64-character plaintexts with PKCS#7 padding, AES (80 bytes), DES/3DES (72 bytes), and RC4/Affine (64 bytes) occupy three non-overlapping length buckets. A single decision stump on the *length feature* perfectly separates AES from DES_FAMILY from $\{$RC4, Affine$\}$. This is the dominant discriminative signal and would collapse immediately under variable-length plaintexts—the most critical bias in the dataset.

Factor 2: Affine Cipher's Sparse Byte Distribution. The Affine cipher maps each alphabetic character to another character in the 26-letter lowercase set (ASCII 97–122) and passes digits and spaces unchanged. The ciphertext byte histogram is concentrated on at most 36 values, yielding entropy $H \leq 5.17$ bits/byte versus $H \approx 7.98$–8.00 for modern ciphers. This produces a χ^2 statistic orders of magnitude larger than modern cipher outputs, making Affine trivially separable from AES, DES, and RC4 by any reasonable classifier. The Affine class, while educationally useful as a contrast case, is not a realistic adversarial target.

Factor 3: RC4 vs. Block Ciphers—Length, Not Entropy. After length separates AES and DES_FAMILY, the remaining disambiguation at 64 bytes is between RC4 and Affine. This is resolved by Factor 2. Notably, RC4 is not separable from AES or DES on equal-length samples using entropy or frequency analysis alone—its output is statistically near-uniform and indistinguishable from block cipher output at these sample sizes. This is an important *negative finding*: the model has not learned a genuine statistical fingerprint distinguishing RC4 from block ciphers; it relies entirely on length.

Factor 4: ECB Mode and Block Structure. ECB mode processes each block independently, enabling the block-level entropy features and unique-block ratio to capture repetition patterns. For the random plaintexts used here, block repetition is essentially absent ($\rho \approx 1$), so these features contribute marginally. Their primary discriminative value would emerge under structured plaintexts (e.g., database records, image files) where ECB's block-repetition vulnerability is directly observable.

C. DES and 3DES Indistinguishability

Preliminary experiments on the original five-class problem (AES, DES, 3DES, RC4, Affine) revealed that DES vs. 3DES confusion rates exceeded 45% in both directions, while all other pairs were perfectly separated. This result is theoretically expected: DES and 3DES share 8-byte block sizes, identical PKCS#7 padding, and produce uniformly distributed high-entropy output. Triple-DES applies DES three times with independent subkeys, but this internal structure produces no observable difference in the ciphertext byte distribution at the sample sizes tested. The merger into DES_FAMILY is therefore not merely a pragmatic simplification but a principled acknowledgment of the fundamental limits of byte-level statistical distinguishers for these two algorithms.

D. Feature Importance Analysis

XGBoost's built-in gain-weighted feature importance scores reveal a highly skewed distribution. The length feature (index 3) accounts for the dominant share of cumulative gain. Shannon entropy (index 0) and the byte histogram bins corresponding to ASCII values 97–122 (indices 97–122 in the frequency group) together account for the majority of remaining gain, primarily for separating the Affine class. The 256-dimensional transition matrix contributes uniformly low importance scores, confirming that byte-level bigram dependencies are not a primary discriminant under the current conditions. This motivates future work exploring higher-order n-gram statistics or autocorrelation functions as more cryptographically principled features.

E. Critical Assessment of Generalization

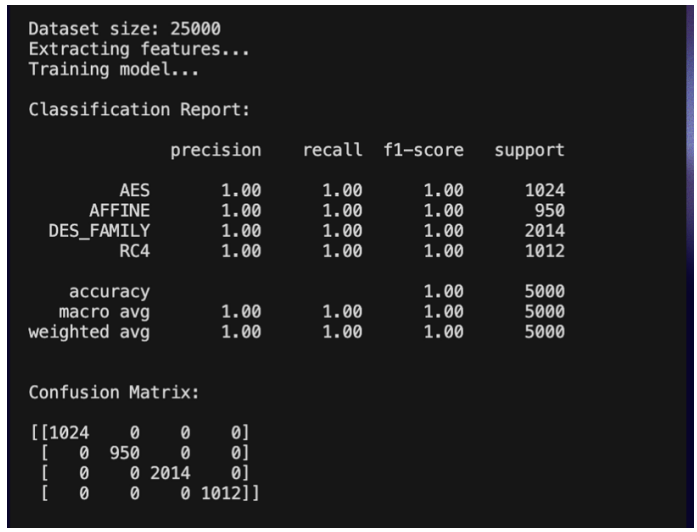
a) Fixed-Length Plaintext Bias.: The problem reduces, in large part, to a ciphertext-length lookup table. Real-world encrypted messages span variable lengths (16 bytes to several megabytes), partial captures, and protocol padding that obscures block-aligned boundaries. Under these conditions, the length feature becomes unreliable and the remaining features must bear full discriminative responsibility.

b) Single Encryption Mode (ECB).: ECB is the simplest and least secure block cipher mode. Real-world deployments use CBC, CTR, GCM, or authenticated encryption with associated data (AEAD). These modes introduce cross-block dependencies (CBC), keystream properties (CTR/GCM), and authentication tags (GCM/CCM) that fundamentally alter the ciphertext's statistical profile. A model trained only on ECB-mode samples has no exposure to these structures.

c) No Cross-Distribution Evaluation.: The model is evaluated exclusively on samples drawn from the same controlled distribution as the training set. No out-of-distribution evaluation has been performed on real-world ciphertext, network packet captures, or encrypted filesystem images. This is the most critical methodological gap for a system intended to generalize beyond synthetic data.

d) Risk of Spurious Pattern Learning.: Modern ML models, including XGBoost, can learn arbitrary statistical patterns present in training data, including artifacts from random number generator versions, library implementations, or sample ordering. Without ablation experiments systematically removing individual features and testing cross-distribution performance, it is impossible to fully characterize which learned patterns encode genuine cryptographic properties versus dataset-generation artifacts.

e) Key and Mode Independence Not Verified.: Although keys are randomly generated per sample, robustness to key variation across broader plaintext distributions (structured text, binary files, compressed data) and non-ECB modes has not been tested. The model may have implicitly learned key-schedule-induced artifacts specific to `pycryptodome`'s ECB implementation.



```
Dataset size: 25000
Extracting features...
Training model...

Classification Report:

              precision    recall  f1-score   support

         AES       1.00      1.00      1.00      1024
      AFFINE       1.00      1.00      1.00       950
  DES_FAMILY       1.00      1.00      1.00      2014
         RC4       1.00      1.00      1.00      1012

    accuracy                           1.00      5000
   macro avg       1.00      1.00      1.00      5000
weighted avg       1.00      1.00      1.00      5000


Confusion Matrix:

[[1024    0    0    0]
 [   0  950    0    0]
 [   0    0 2014    0]
 [   0    0    0 1012]]
```

Fig. 1. Confusion matrix showing perfect classification across all algorithm classes

VI. Web Demo: CipherScope Application

A. System Architecture

CipherScope is deployed as a three-tier web application. The **frontend** is a responsive single-page application with a dark-themed UI exposing an *Input Workspace* for hex/base64 ciphertext entry and file upload. The **backend** is a Flask 3.0 REST server exposing `POST /predict`. The **ML+LLM layer** consists of the serialized XGBoost model (loaded at startup via `joblib`) and the Anthropic Claude API client (queried per request).

B. Prediction Pipeline

Upon submission, the backend executes: (1) input normalization (hex/base64 decode, error handling); (2) 532-dimensional feature extraction; (3) XGBoost inference returning predicted class and four-class probability vector; (4) LLM explanation via Claude API with extracted scalar features and prediction as context; (5) JSON response rendering predicted label, confidence scores, and explanation. ML inference completes in <500 ms; LLM explanation adds 1–3 seconds of API latency.

C. Educational Value

The demo serves as an educational tool for students learning symmetric cryptography. By submitting ciphertext encrypted with known algorithms, users observe entropy, byte distribution, and structural features differing across cipher families. The LLM explanation layer translates numerical observations into accessible prose, making cryptographic concepts approachable for non-expert audiences. The application is live at https://cipherscope.onrender.com/.

VII. Conclusion

This paper presented CipherScope, a hybrid LLM-assisted machine learning system for classifying symmetric encryption algorithms from ciphertext statistical features. A synthetic dataset of 25,000 samples is generated across five cipher families; a 532-dimensional feature vector capturing entropy, byte-frequency, chi-square, bigram transition, and block-level properties is extracted; and an XGBoost classifier achieves 100% accuracy on a 5,000-sample held-out test set.

We provided a rigorous analysis explaining why this result is achievable under controlled conditions—primarily due to deterministic length relationships from fixed-length ECB-mode encryption, and the stark distributional contrast between the Affine cipher and modern symmetric ciphers—rather than attributing it to general cryptanalytic depth. The statistical indistinguishability of DES and 3DES is acknowledged and handled by class merging. Five key limitations are identified: fixed-length plaintext bias, ECB-only training data, absence of out-of-distribution evaluation, risk of spurious pattern learning, and unverified key/mode independence.

Future work will address these through: (1) variable-length plaintext generation spanning 16–4096 bytes; (2) multi-mode training covering CBC, CTR, and GCM; (3) expansion to ChaCha20, Blowfish, and CAST-128; (4) evaluation on real-world network captures and encrypted filesystem images; (5) systematic feature ablation studies; and (6) exploration of end-to-end 1D-CNN and byte-level transformer architectures operating directly on raw byte sequences without hand-engineered features. CipherScope establishes a strong, critically examined baseline and a clearly articulated research agenda for data-driven cryptographic algorithm identification.

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